

RETAIL CUSTOMER ANALYTICS

Consumer Shopping Trends

Analysis & Recommendations for Marketing and Product Strategy

Prepared by: Namit

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Data source: cleaned_customer_behavior.csv (3,900 transactions)

Tools: Python (pandas), Microsoft SQL Server (T-SQL), Power BI

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1. Executive Summary

This report analyses 3,900 retail transactions to answer the company's central question: how can consumer shopping data be used to identify trends, improve customer engagement, and optimize marketing and product strategy? The underlying data was cleaned and feature-engineered in Python, then queried through 15 structured business questions in T-SQL, spanning revenue patterns, product and category performance, customer segmentation, and behavioral cross-factors. The findings below are drawn directly from that analysis, every figure in this report is a computed result, not an estimate.

Headline Finding

The subscription and promotion program has zero reach with women.

0% of female customers (0 of 1,248) are subscribed or have ever used a discount/promo code, compared with 39.7% and 63.2% of male customers respectively, despite average order values being virtually identical (\$60.25 for women vs. \$59.54 for men). This is the single largest, cleanest growth opportunity identified in the data.

Other Key Findings

- Middle-aged customers (41–60) generate 38.4% of total revenue (\$89,445 of \$233,081), more than double the young-adult (18–25) contribution of 14.9%.
- 79.9% of customers already qualify as “Loyal” (11+ prior purchases), yet 72.4% of repeat buyers (more than 5 prior orders, 3,476 customers) are not subscribed, leaving 2,518 proven customers with no recurring-revenue relationship.
- Clothing is the top-revenue category in every season, and total seasonal revenue varies by less than 8%, demand in this dataset is stable year-round rather than sharply seasonal.
- Discount usage has no measurable effect on customer satisfaction (3.74 average rating with a discount vs. 3.76 without), discounting can be used as an acquisition lever without a satisfaction trade-off, though three products (Hat, Sneakers, Coat) are discounted on roughly half of their sales and merit a margin review.
- Shipping type, payment method, and purchase frequency were all tested as potential spend drivers; none produced more than a 3.5% difference in average order value and are deprioritized as primary levers.

Top Recommendations

1. Open the subscription and promotion program to female customers, the largest identified whitespace.
2. Concentrate marketing investment on the 41–60 age cohort, which already drives more revenue than any other segment.
3. Run a loyalty-to-subscription campaign targeted at the 2,518 unsubscribed repeat buyers.

4. Maintain Clothing as the year-round core assortment rather than rotating merchandising by season.
5. Audit margins on Hat, Sneakers, and Coat given their high discount incidence, while continuing to use discounts broadly as an acquisition tool.

2. Business Context & Objectives

The company has observed shifting purchase patterns across customer demographics, product categories, and sales channels, but lacks a clear, data-backed view of which factors, discounts, reviews, seasonality, payment method, or demographic segment, actually drive customer decisions and repeat purchases. Management commissioned this analysis to translate the company's consumer shopping trends dataset into a concrete, evidence-based answer to a single guiding question:

Guiding Business Question

“How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?”

This report addresses that question in four parts: a description of the data and how it was prepared (Section 3–4), a structured set of findings organized around revenue, product performance, customer loyalty, and behavior (Section 5), a prioritized set of recommendations (Section 6), and an honest accounting of the analysis's limitations (Section 7).

3. Methodology

The analysis followed a three-stage pipeline, moving from raw transactions to a decision-ready set of recommendations.

3.1 Data Cleaning & Preparation (Python)

The raw dataset (3,900 rows, 18 columns) was profiled, cleaned, and feature-engineered in a Jupyter notebook using pandas. Specific steps:

- Missing values: 37 missing review_rating values (0.9% of rows) were imputed using the median rating within each row's own product category, rather than a single global median, so imputed values stay consistent with how that category is typically rated.
- Column standardization: all column headers were normalized to lowercase snake_case (e.g., "Purchase Amount (USD)" → purchase_amount_usd) for direct use in both pandas and SQL.
- Feature engineering: age_group buckets the continuous age column into four labeled cohorts (young_adult 18–25, adult 26–40, middle_aged 41–60, senior 61+); purchase_frequency_days converts the categorical frequency_of_purchases field (e.g., "Weekly," "Annually") into an approximate numeric day count.
- Redundancy check: discount_applied and promo_code_used were confirmed identical across all 3,900 rows and promo_code_used was dropped as redundant.

The cleaned file (19 columns, zero missing values) was exported as the single source of truth for the SQL analysis phase.

3.2 Structured Analysis (SQL)

15 business questions were written in T-SQL against the cleaned dataset, organized into four sections: revenue and spending patterns, product and category performance, customer segmentation and loyalty, and behavioral cross-factor analysis. Each query was designed around a specific business question and a stated rationale for why that framing was chosen (see Appendix B for the full list).

3.3 Synthesis (This Report)

Query outputs were reviewed collectively, including cross-referencing results across queries, such as subscription and discount usage by gender, to identify the handful of findings with the clearest business implications, each paired with a concrete recommendation.

4. Dataset Overview

cleaned_customer_behavior.csv is the single source of truth for every figure in this report.

Metric	Value
Transactions analyzed	3,900
Columns (post-cleaning)	19 (2 engineered)
Total revenue captured	\$233,081
Average review rating	3.75 / 5
Missing values remaining	0
States represented	50
Product categories	4 (Clothing, Accessories, Footwear, Outerwear)
Distinct products	25
Overall subscription rate	27.0%
Overall discount-usage rate	43.0%

4.1 Column Reference

Column	Description
customer_id	Unique customer / transaction identifier
age, age_group	Customer age; engineered 4-bucket cohort (young_adult, adult, middle_aged, senior)
gender	Male / Female
item_purchased, category	Product name and its category (Clothing, Accessories, Footwear, Outerwear)
purchase_amount_usd	Order value in USD
location	US state
size, color, season	Product attributes and the season of purchase
review_rating	1–5 product rating (37 values imputed by category median)
subscription_status	Whether the customer is enrolled in the subscription program
shipping_type	Standard, Express, Free Shipping, Next Day Air, etc.
discount_applied	Whether a discount or promo code was used
previous_purchases	Count of prior orders ,the basis for the loyalty segmentation in Section 5.3
payment_method	Card, digital wallet, cash, or bank transfer
frequency_of_purchases, purchase_frequency_days	Original categorical cadence label and its engineered numeric day-count equivalent

5. Detailed Findings

Findings are grouped into the same four sections used in the SQL analysis. Each subsection states the business question, the result, and its interpretation.

5.1 Revenue & Spending Patterns

Revenue by Gender

Gender	Total Revenue	Share of Revenue	Avg Order Value
Male	\$157,890	67.7%	\$59.54
Female	\$75,191	32.3%	\$60.25

Male customers account for two-thirds of revenue, but this tracks their share of the customer base (2,652 of 3,900 customers, or 68.0%) rather than a difference in typical spend ,average order value is nearly identical between genders. Gender alone is a volume signal here, not a basket-size signal. The far more consequential gender-related pattern is described in the Headline Finding in Section 1 and detailed in Section 5.3.

Discount Users Spending Above Average

839 of 3,900 customers (21.5%) used a discount and still spent at or above the dataset average of \$59.76 per order. This segment is not purely price-sensitive, they respond to a discount offer but are not solely driven by it, making them strong candidates for loyalty or lower-discount-depth offers rather than continued deep discounting.

Shipping Type and Order Value

Shipping Type	Avg Order Value
Express	\$60.48
Standard	\$58.46

Express shipping customers spend about 3.5% more per order than Standard shipping customers ,a real but modest signal that willingness to pay for speed correlates weakly with basket size.

Revenue by Age Group

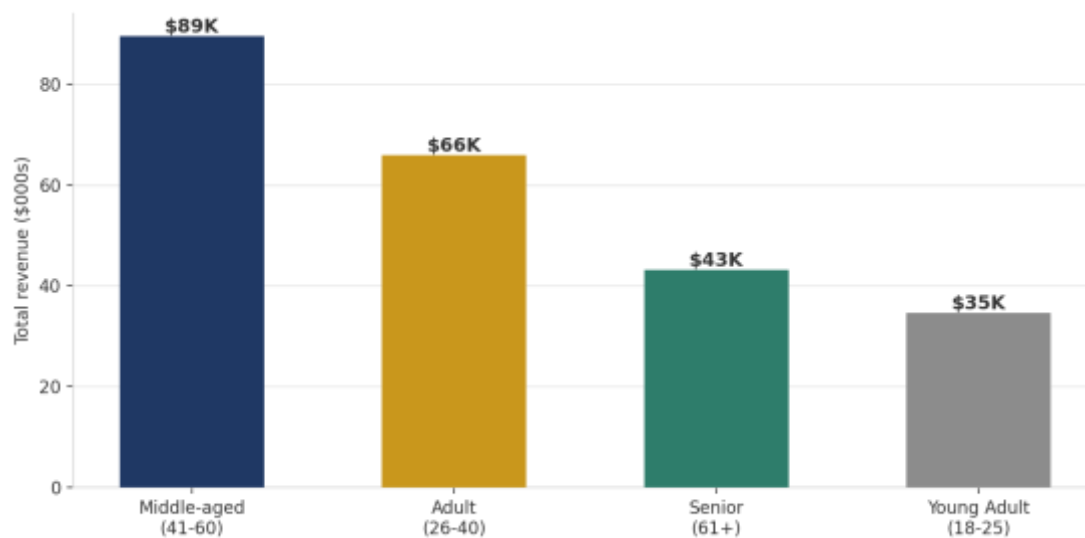


Figure 1. Total revenue by engineered age cohort.

Age Group	Total Revenue	Share of Revenue
Middle-aged (41–60)	\$89,445	38.4%
Adult (26–40)	\$65,842	28.2%
Senior (61+)	\$43,164	18.5%
Young Adult (18–25)	\$34,630	14.9%

The middle-aged cohort is the single largest revenue driver by a wide margin, generating 2.6× the revenue of young adults. This concentration is a clear signal for where marketing reach and retention investment have the largest ceiling.

Top 10 Locations by Average Revenue per Customer

Averaging revenue per customer (rather than totaling it) controls for population size and surfaces states with genuinely higher-value customers rather than simply more of them.

Location	Customers	Avg Revenue / Customer	Total Revenue
Alaska	72	\$67.60	\$4,867
Pennsylvania	74	\$66.57	\$4,926
Arizona	65	\$66.55	\$4,326
West Virginia	81	\$63.88	\$5,174
Nevada	87	\$63.38	\$5,514
Washington	73	\$63.33	\$4,623
North Dakota	83	\$62.89	\$5,220
Virginia	77	\$62.88	\$4,842
Utah	71	\$62.58	\$4,443
Michigan	73	\$62.10	\$4,533

The spread across the top 10 locations is modest (Alaska's \$67.60 vs. Michigan's \$62.10 ,about 9%), and each state represents a small sample (65–87 customers). Location is a secondary lever at best relative to age and the gender-access gap.

5.2 Product & Category Performance

Top-Rated and Most-Discounted Products

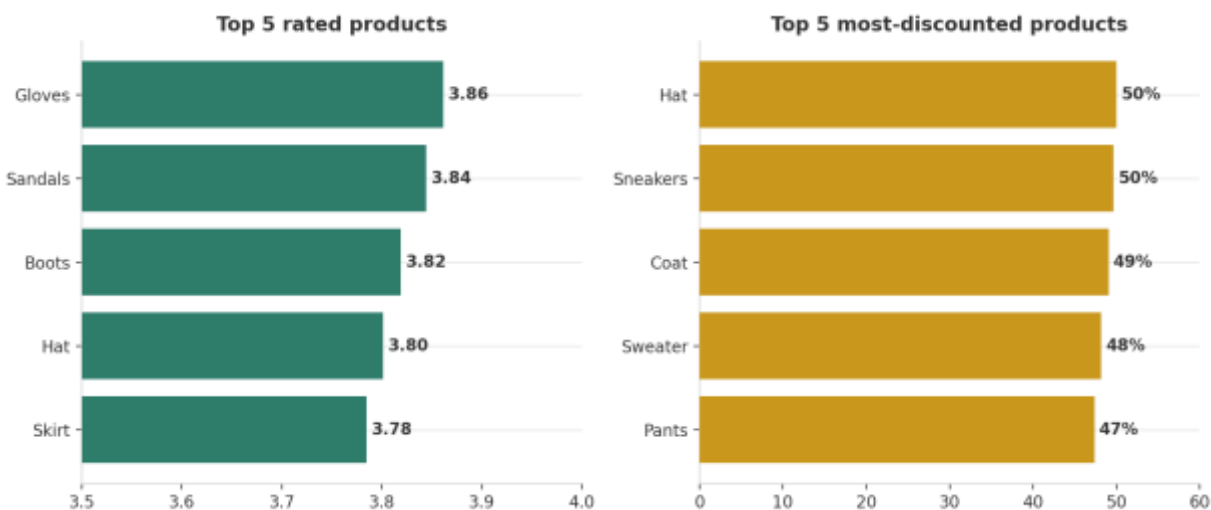


Figure 2. Top 5 products by average review rating (left) and by discount incidence (right).

Rank	Top Rated Product	Avg Rating	Most Discounted Product	Discount Rate
1	Gloves	3.86	Hat	50.0%
2	Sandals	3.84	Sneakers	49.7%
3	Boots	3.82	Coat	49.1%
4	Hat	3.80	Sweater	48.2%
5	Skirt	3.78	Pants	47.4%

Hat appears on both lists: it is well-rated (3.80, 4th overall) but discounted on exactly half of its sales. Combined with Sneakers and Coat, all three of the most-discounted products sit close to a 50% discount incidence, which warrants a margin review even though satisfaction is unaffected (see Section 5.4).

Top 3 Products per Category

Category	Rank 1	Rank 2	Rank 3
Accessories	Jewelry (171 orders)	Belt (161 orders)	Sunglasses (161 orders)
Clothing	Blouse (171 orders)	Pants (171 orders)	Shirt (169 orders)
Footwear	Sandals (160 orders)	Shoes (150 orders)	Sneakers (145 orders)
Outerwear	Jacket (163 orders)	Coat (161 orders)	—

Seasonal Category Performance

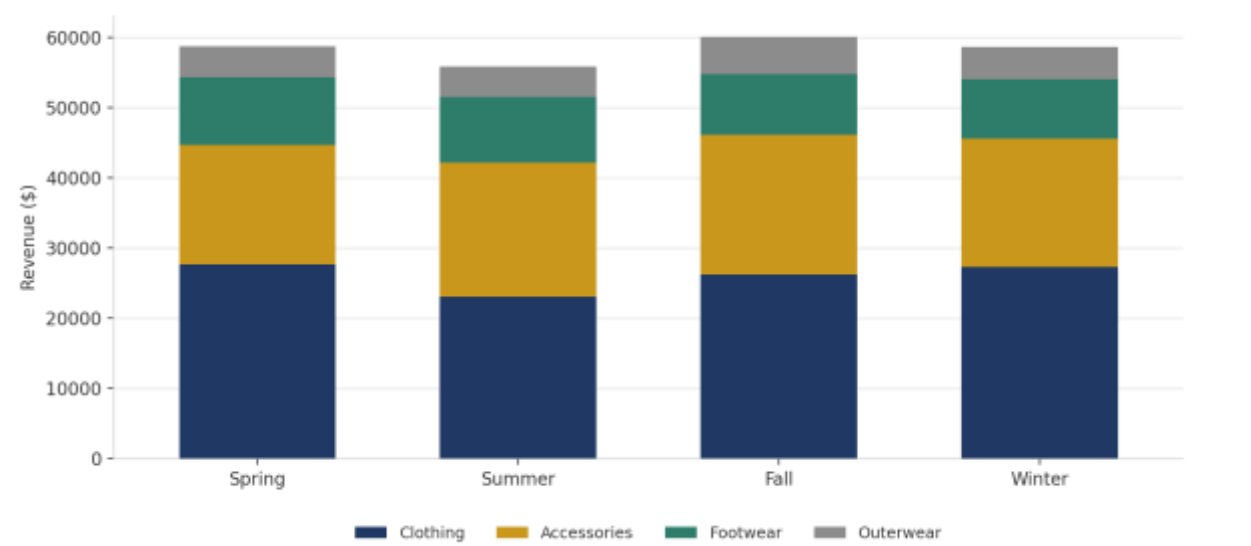


Figure 3. Revenue by category within each season.

Season	Total Revenue	Leading Category
Fall	\$60,018	Clothing (\$26,220)
Spring	\$58,679	Clothing (\$27,692)
Winter	\$58,607	Clothing (\$27,274)
Summer	\$55,777	Clothing (\$23,078)

Clothing is the top-revenue category in every season with no exceptions, and total seasonal revenue varies by less than 8% peak to trough (Fall \$60,018 vs. Summer \$55,777). Outerwear is the one category with a visible seasonal pattern, peaking in Fall just ahead of Winter. Overall, demand in this dataset is stable year-round rather than sharply seasonal, arguing against season-specific merchandising rotation in favor of consistent, year-round investment in the Clothing assortment.

5.3 Customer Segmentation & Loyalty

The Subscription and Discount Gender Gap

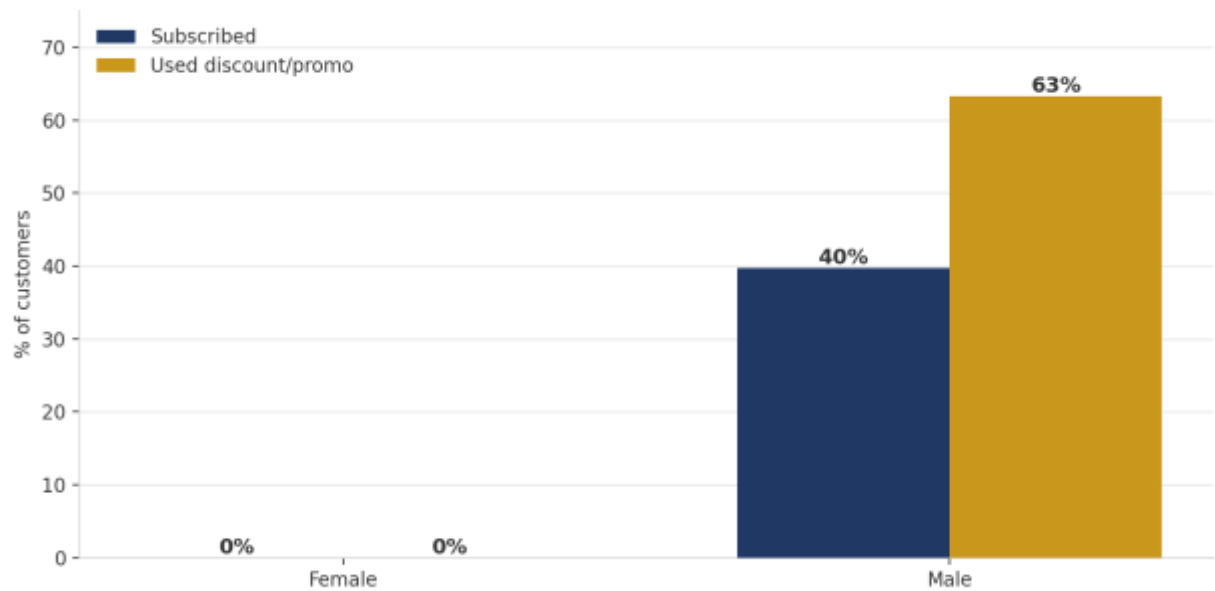


Figure 4. Subscription and discount/promo usage rate by gender.

Gender	Subscribed	Used a Discount / Promo Code
Female	0.0% (0 of 1,248)	0.0% (0 of 1,248)
Male	39.7% (1,053 of 2,652)	63.2% (1,677 of 2,652)

This is the most striking pattern in the dataset: every one of the 1,053 subscribers and every one of the 1,677 customers who ever used a discount or promo code is male. No female customer in this dataset has ever subscribed or received a discount, despite average order values being virtually identical between genders. Whether this reflects a genuine program gap or a data-collection artifact in the source system, it is the clearest, single largest opportunity surfaced by this analysis and is flagged as a limitation to validate in Section 7.

Subscription Status and Spend

Subscription Status	Customers	Avg Spend	Total Revenue
Not Subscribed	2,847	\$59.87	\$170,436
Subscribed	1,053	\$59.49	\$62,645

Subscribed and non-subscribed customers spend almost identically per order (\$59.49 vs. \$59.87), the subscription program's value lies in retention and repeat engagement, not in a higher per-order basket size.

Customer Loyalty Segmentation

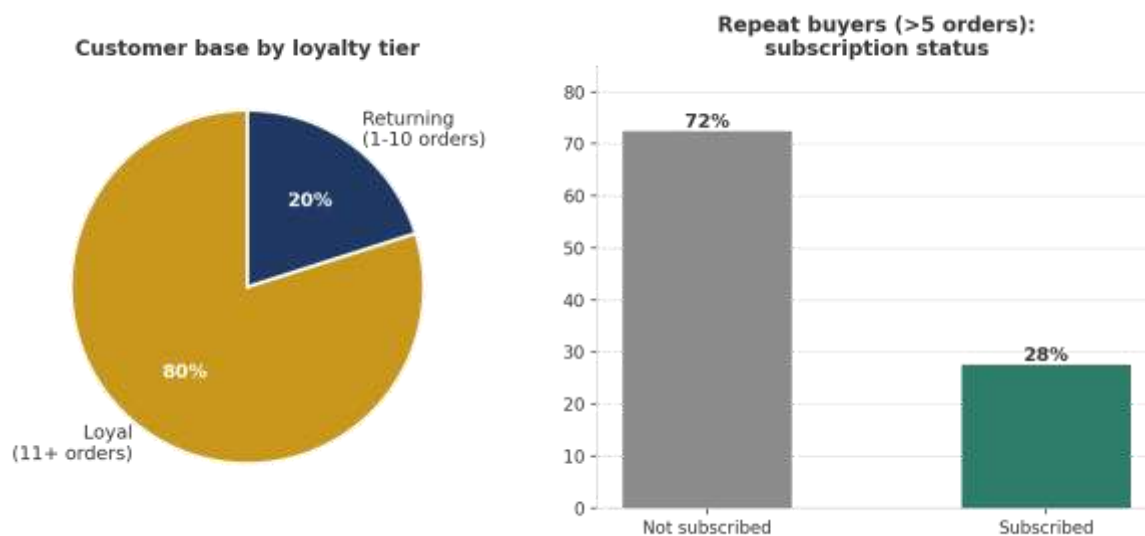


Figure 5. Customer base by loyalty tier (left) and subscription status among repeat buyers (right).

Segment	Definition	Customers	Share
Loyal	11+ previous purchases	3,116	79.9%
Returning	1–10 previous purchases	784	20.1%

No customer in this dataset shows zero previous purchases, there is no true “New” (first-time) segment. This dataset captures an already-established customer base, not an acquisition funnel; see Section 7 for the implication.

Repeat Buyers and Subscription Status

Subscription Status	Repeat Buyers (>5 orders)	Share
Not Subscribed	2,518	72.4%
Subscribed	958	27.6%

Of the 3,476 customers with more than 5 previous purchases, proven, established buyers, nearly three-quarters (2,518) are not subscribed. This is a large, low-risk upsell target: these customers have already demonstrated repeat purchase behavior without a subscription.

Purchase Frequency and Spend

Purchase Cadence	Days Between Purchases	Customers	Avg Order Value	Avg Previous Purchases
Weekly	7	539	\$58.97	25.8
Bi-Weekly	14	547	\$60.69	24.8
Fortnightly	14	542	\$59.05	25.3
Monthly	30	553	\$59.33	25.3
Quarterly	90	563	\$59.98	26.9
Every 3 Months	90	584	\$60.08	25.0

Purchase Cadence	Days Between Purchases	Customers	Avg Order Value	Avg Previous Purchases
Annually	365	572	\$60.17	24.6

Contrary to the common retail hypothesis that frequent buyers have smaller baskets but higher lifetime value, average order value is essentially flat across every purchase-cadence cohort (\$58.97–\$60.69, under 3% spread), and average previous-purchase count is similarly flat (~25 regardless of cadence). Purchase frequency does not predict basket size or lifetime order count in this dataset.

5.4 Behavioral & Cross-Factor Analysis

Discount Usage and Review Ratings

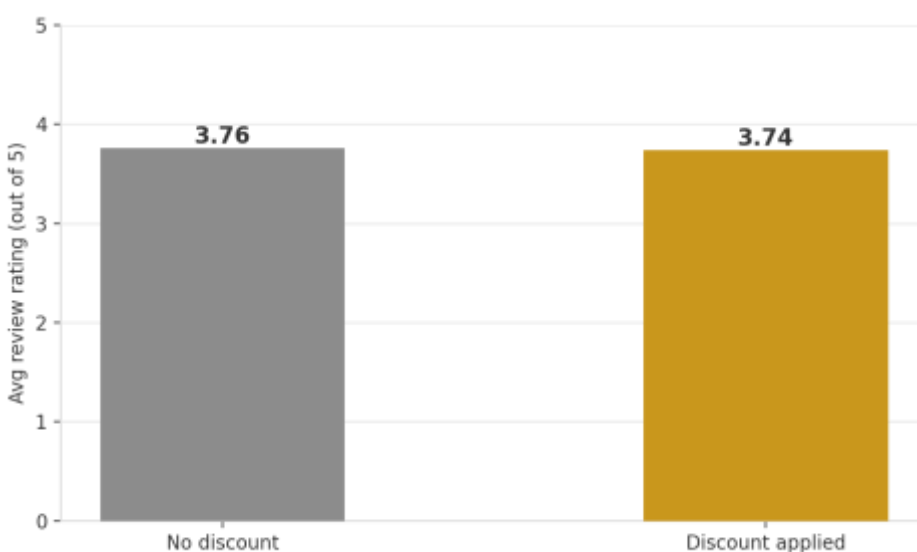


Figure 6. Average review rating by discount usage.

Discount Applied	Purchases	Avg Review Rating
No	2,223	3.76
Yes	1,677	3.74

Review ratings are effectively unchanged by discount usage (3.76 vs. 3.74), customers do not rate discounted purchases meaningfully lower or higher than full-price ones. Discounting can be used as a volume and acquisition lever without a satisfaction trade-off, though the margin implications of heavy discounting on specific products (Section 5.2) still warrant review.

Payment Method and Order Value



Figure 7. Average order value by shipping type, payment method, and purchase frequency ,all narrow, secondary factors.

Payment Method	Transactions	Avg Order Value
Debit Card	636	\$60.92
Credit Card	671	\$60.07
Bank Transfer	612	\$59.71
Cash	670	\$59.70
PayPal	677	\$59.25
Venmo	634	\$58.95

Payment method shows a narrow ~3% spread between the highest (Debit Card, \$60.92) and lowest (Venmo, \$58.95) average order values, no method stands out as a meaningful spend driver, and payment method should be deprioritized as a targeting variable relative to age, loyalty status, and the gender-access gap.

6. Strategic Recommendations

The following recommendations are ordered by revenue opportunity and strength of evidence.

6.1 Open the Subscription & Promotion Program to Women

Zero female customers currently subscribe or receive discounts, despite spending on par with men (\$60.25 vs. \$59.54 average order value). This is the single largest untapped growth lever identified in this analysis. Recommended first step: pilot a subscription/promo invitation to a representative sample of female customers and measure enrollment and spend lift before a full rollout (see Section 7 for a data-validation caveat on this finding).

6.2 Concentrate Marketing on the 41–60 Age Cohort

This group already drives 38.4% of revenue, more than the adult and senior cohorts combined. Prioritize retention and cross-sell campaigns for this segment before expanding acquisition spend into lower-revenue cohorts.

6.3 Convert Loyal Repeat Buyers into Subscribers

2,518 customers with more than 5 previous orders are not subscribed. A targeted loyalty-to-subscription upgrade campaign converts already-proven buyers into a recurring-revenue relationship at comparatively low acquisition risk.

6.4 Keep Clothing as the Year-Round Anchor Category

Clothing leads every season by a wide margin, and total seasonal revenue is stable (under 8% variation). Invest in depth and consistency of the Clothing assortment rather than season-specific merchandising rotation; reserve seasonal merchandising emphasis for Outerwear, the one category with a clear seasonal pattern (peaking in Fall).

6.5 Use Discounts Freely, but Audit Margins on Hat, Sneakers & Coat

Review ratings are unaffected by discounting (3.76 vs. 3.74), so discounts remain a safe acquisition and volume tool broadly. However, Hat, Sneakers, and Coat are each discounted on roughly half of their sales, a margin review on these three items is warranted before continuing that discount depth indefinitely.

7. Limitations & Next Steps

7.1 Limitations

- No customer in this dataset shows zero previous purchases, it reflects an already-established base, not first-time buyers. First-purchase conversion and acquisition-funnel questions cannot be analyzed until that cohort is tracked and included.
- Subscription and discount fields are populated only for male records in the source data. Before finalizing the female-targeting business case (Recommendation 6.1), this pattern should be validated directly with the data source or system of record to rule out a collection or extraction artifact rather than a genuine program gap.
- Findings reflect a single historical snapshot of 3,900 orders. Revenue concentrations and behavioral patterns should be re-checked as new transactions accumulate, rather than treated as permanent.
- Location-level findings (Section 5.1) are based on small per-state samples (65–87 customers) and should be treated as directional, not statistically definitive.

7.2 Next Steps

1. Build and maintain the Power BI dashboard (in progress) so the views in this report refresh automatically each reporting cycle.
2. Validate the gender / subscription-field pattern with the source system before scaling the female-targeting pilot.
3. Pilot the subscription/promo invitation to a sample of female customers and measure enrollment and spend lift.
4. Instrument true first-purchase tracking so future analyses can address acquisition-funnel questions, not just retention and loyalty questions.
5. Re-run this analysis on a recurring (e.g., quarterly) basis as new transactions accumulate.

Appendix A: Full Column Reference

All 19 columns in cleaned_customer_behavior.csv, the source file for every query in this report:

#	Column	Type
1	customer_id	Integer identifier
2	age	Integer (18–70)
3	gender	Male / Female
4	item_purchased	Text (25 distinct products)
5	category	Text (4 categories)
6	purchase_amount_usd	Numeric (\$20–\$100)
7	location	Text (50 US states)
8	size	Text (S/M/L/XL)
9	color	Text (25 distinct colors)
10	season	Text (4 seasons)
11	review_rating	Numeric (2.5–5.0, 37 imputed)
12	subscription_status	Yes / No
13	shipping_type	Text (5 methods)
14	discount_applied	Yes / No
15	previous_purchases	Integer (1–50)
16	payment_method	Text (6 methods)
17	frequency_of_purchases	Text (7 categories)
18	age_group	Engineered: young_adult / adult / middle_aged / senior
19	purchase_frequency_days	Engineered: numeric day-count equivalent of frequency_of_purchases

Appendix B: SQL Business Questions Reference

The 15 T-SQL business questions analyzed against the cleaned dataset (full query text is available in `customer_shopping_trend_analysis.sql` in the project repository).

Section 1: Revenue & Spending Patterns

1. What is the total revenue generated by male vs. female customers?
2. Which customers used a discount but still spent more than the average purchase amount?
3. Compare the average purchase amount between Standard and Express shipping.
4. What is the revenue contribution of each age group?
5. Which 10 locations generate the highest average revenue per customer?

Section 2: Product & Category Performance

1. Which are the top 5 products with the highest average review rating?
2. Which 5 products have the highest percentage of purchases made with a discount applied?
3. What are the top 3 most purchased products within each category?
4. Which category generates the most revenue in each season, and how does total revenue vary by season overall?

Section 3: Customer Segmentation & Loyalty

1. Do subscribed customers spend more than non-subscribers?
2. Segment customers into New, Returning, and Loyal based on previous purchases, how many fall into each segment?
3. Are repeat buyers (more than 5 previous purchases) also likely to subscribe?
4. How does purchase frequency relate to spend, do more frequent buyers spend less per transaction but more over their history?

Section 4: Behavioral & Cross-Factor Analysis

1. Is there a relationship between discount usage and review rating?
2. Which payment method is associated with the highest average purchase amount?